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CloudNativeCon

Europe 2026

#KubeCon #CloudNativeCon

Longhorn

Intro, Deep Dive and Q&A

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Agenda

- Longhorn Overview
- Project and Release Status
- Data Engines, and Performance
- 1.11 Key Features
- 1.12 Roadmap
- Community Highlights

Longhorn Overview



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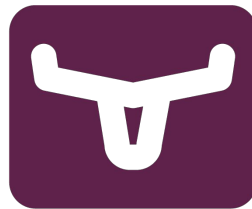


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- Cloud Native “Distributed Block Storage” for Kubernetes
- Highly Available Persistent Volume
- Storage Space Efficiency
- Data Durability, Availability, Consistency, Protection (Snapshot, Backup), Recovery, Safety
- Advanced Volume Operations
- Operation Simplicity & Reliability
- Deployment Platform Agnostic
- Virtualization Workload Support, KubeVirt

👉 <https://longhorn.io/docs/latest/>



LONGHORN

CNCF **Incubating** Project

Longhorn Architecture

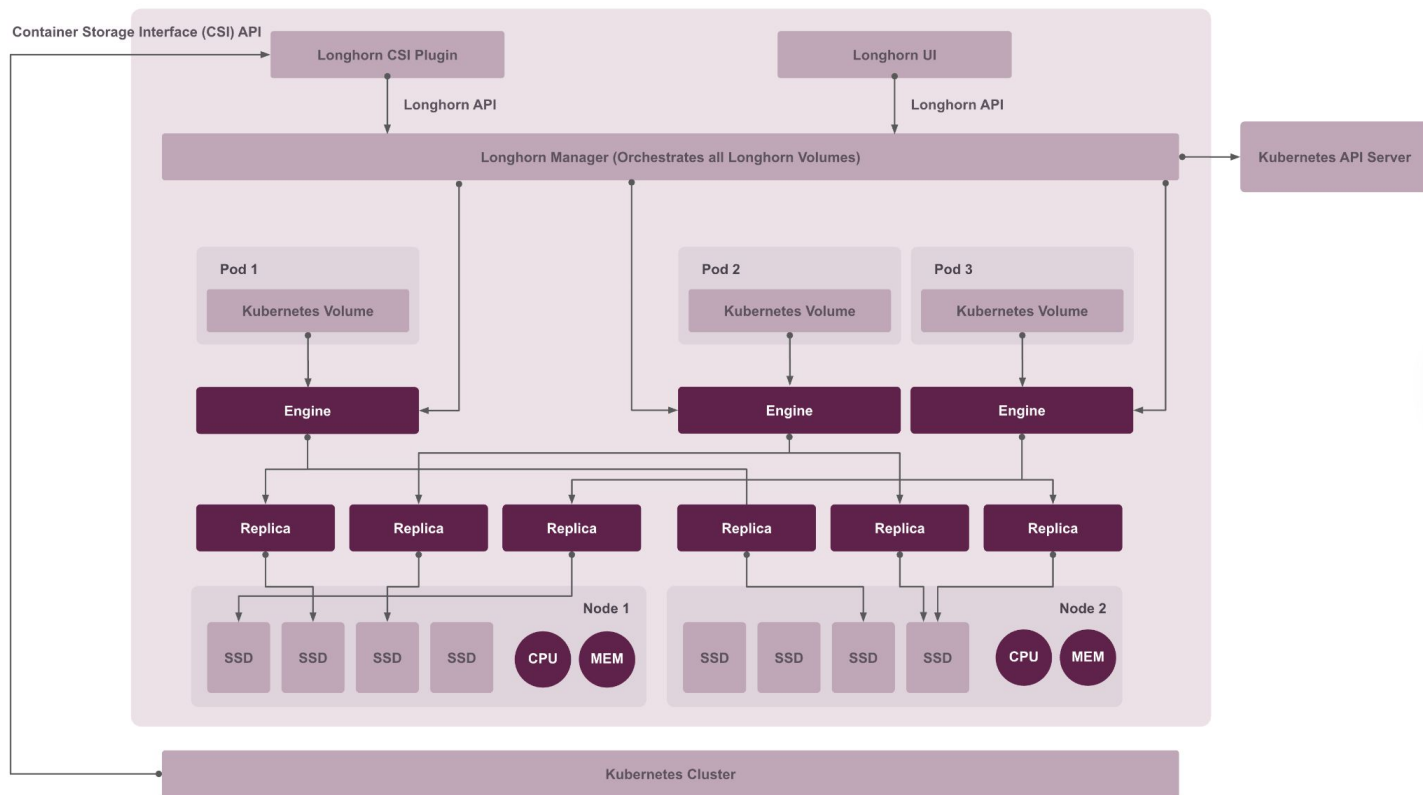


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Longhorn Capacities



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- **Kubernetes Persistent Volume Support**

- Block, FS volumes
- RWO, RWX, RWX (block with migration capacity), RWOP

- **CSI Protocol Support**

- Volume Provision, Attachment, Snapshot, Clone, Expansion

- **Volume Capabilities**

- Thin provisioning
- Snapshot, CoW
- Trim, Expand
- Live upgrade, Live migration, *Engine migration (v2)

- **IO Performance**

- Data Locality, Strict-Local Volume
- V1, V2 Data Engines

- **Intuitive UI & CLI (longhornctl) & Monitoring Metrics**

- **Storage, Storage Topology, Replica Scheduling**

- v1 & v2 Longhorn FS/Block disk managed by SPDK driver
- Disk/Node/Zone replica scheduling anti-affinity
- Storage tag
- CSIStorageCapacity

- **Data Protection**

- Data replication
- Data encryption in traint & at rest
- Bit-rot protection

- **Data Services**

- In-cluster snapshot & revert
- Out-of-cluster backup & restore
- Disaster recovery volume

- **Data Reduction**

- Space efficiency for Snapshot
- Backup compression

Project Status



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Growth Metrics

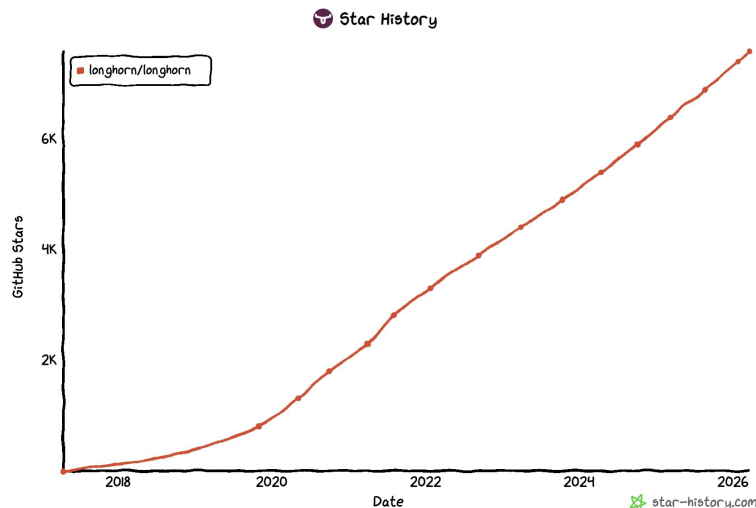
- 7.5K GitHub Stars
- 214K Nodes (🙌 35% from 159K in FY25)
- 46K Clusters (🙌 59% from 29K in FY25)

Platform Agnostic

- On-Prem, Cloud, Edge
- AI, Telco, Data, Virtualization, Observability, and more
- End Users & Service/Solution Providers
- Dynamic and Immutable OSs (e.g., SL Micro, Talos, Google Container-Optimized OS)

Achieving Graduation

- <https://github.com/longhorn/longhorn/blob/master/ADOPTERS.md>
- Community Sustainability



<https://metrics.longhorn.io/>



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210K Installation!

18% continuous growth over the last six months!

Project Status - Cont.



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Utilizing a sprint-based development methodology for release development, community issue prioritization, and QA development.

Open 3 **Closed** 2

Community Review Sprint
Longhorn 2-Week Community Sprint: Starting on Monday of the first week and ending on Sunday of the second week.
#5 updated 39 minutes ago

Longhorn Sprint
Longhorn 2-Week Sprint: Starting on Monday of the first week and ending on Sunday of the second week.
#8 updated 4 hours ago

QA Sprint
Longhorn 2-Week QA Sprint: Starting on Monday of the first week and ending on Sunday of the second week.
#4 updated 2 days ago

Longhorn Sprint

Current Sprint | Next Sprint | Previous Sprint | Previous Sprints | v1.12.0 | v1.11.1 | v1.10.3 | v1.10.2 | All Issues - Board

sprint:@current

Analysis and Design 5

longhorn #1061
[FEATURE] V2 Data Engine Sharding - Experimental
area/v2-data-engine
component/longhorn-instance-manager
experimental
highlight kind/feature priority/0
require/dep require/ui v1.12.0
Sprint 43 34

longhorn #10928
[FEATURE] IPv6 for V2 Data Engine
area/spdk area/v2-data-engine
component/longhorn-instance-manager
highlight kind/feature priority/0
require/auto-e2e-test require/doc
require/dep require/manual-test-plan
v1.12.0 Sprint 43

longhorn #12542
[BUG] lastBackup of a v2 volume may remain empty after a backup is created
area/volume-backup-restore kind/bug
priority/0 reproduce/rare require/backport
require/qa-review-coverage severity/2
v1.12.0 Sprint 43

longhorn #11204
[IMPROVEMENT] Enable the offline-replica-rebuilding setting by default
area/volume-replica-rebuild highlight

Implement 1

longhorn #9205
[BUG] Encrypt volume provided size is 16MB shorter than the claimed size
area/volume-encryption kind/bug priority/0
require/backport require/doc
require/important-note require/dep
require/qa-review-coverage v1.12.0
Sprint 43

Review 1

longhorn #12778
[TASK] Add distro information to upgrade responder requests
area/monitoring kind/task priority/0
v1.12.0 Sprint 43

Ready For Testing

longhorn #12721
[BUG] Longhorn Fails on FIPS enabled systems
area/volume-encryption
require/auto-e2e-test
Sprint 43

longhorn #11442
[FEATURE] Support on checksum calculation
area/cli area/data-in
kind/feature priority/0
require/doc v1.12.0

longhorn #12771
[IMPROVEMENT] Reduce memory usage by optimizing informer caching
area/performance area/scalability back
kind/improvement priority/0
Sprint 43

Project Status - Cont.



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Implement bi-weekly sprint releases to facilitate the evaluation of recently developed features and corresponding fixes.

Tags

v1.12.0-dev-20260322 ...

last week a2502a5 zip tar.gz ± Commits

v1.11.2-dev-20260322 ...

last week 2432147 zip tar.gz ± Commits

v1.11.1 ...

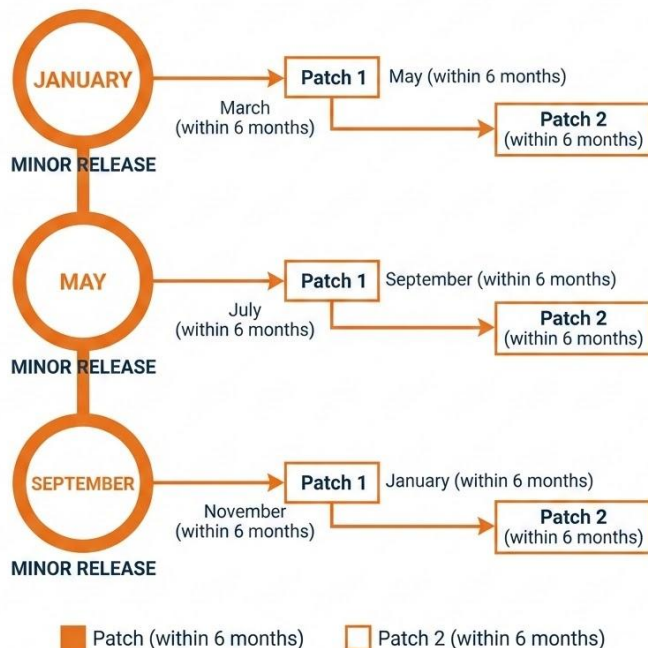
2 weeks ago eb37902 zip tar.gz Notes Downloads

v1.12.0-dev-20260308 ...

2 weeks ago b9962d0 zip tar.gz ± Commits

Release Status

The minor release cycle is quarterly, occurring every four months, with subsequent bimonthly patch releases provided for each minor release over a six-month duration. Minor releases are scheduled for January, May, and September.



Released

- 1.11.0, January 2026
- 1.10.2, January 2026
- 1.11.1, March 2026

Upcoming Releases

- 1.12.0, May 2026
- 1.13.0, September 2026

Longhorn Data Engines



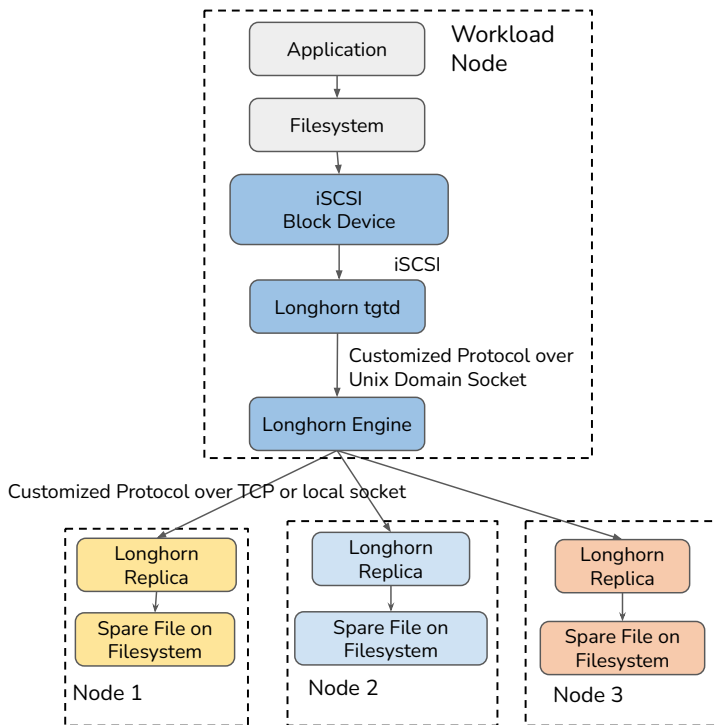
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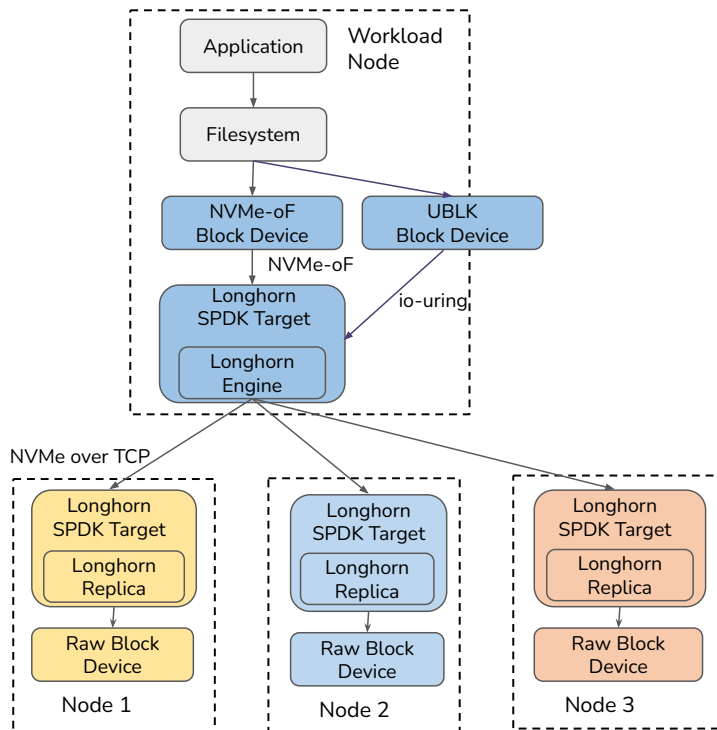
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Longhorn Data Engine v1



Longhorn Data Engine v2



Longhorn V2 Data Engine



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V2 is a high-performance storage architecture replacing the V1 iSCSI model with an SPDK core for higher IOPS and lower latency.

Data Plane (longhorn-spdk-engine, longhorn-instance-manager)

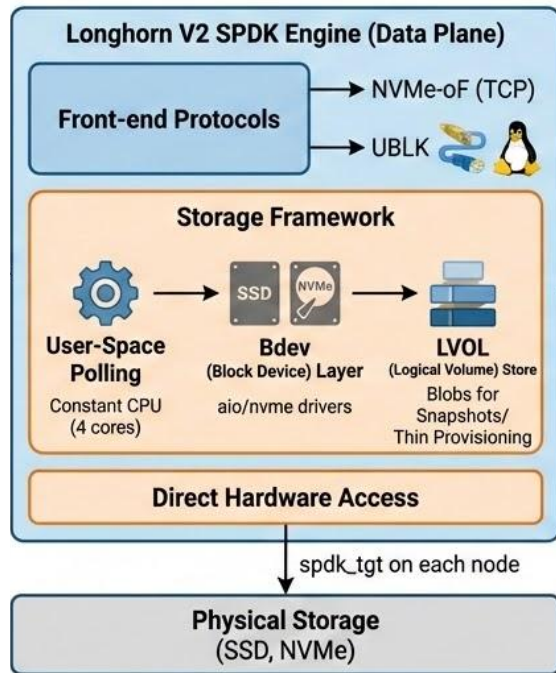
New engine component communicating directly with spdk_tgt via JSON-RPC.

Storage Framework, SPDK

- **User-Space Polling:** Uses a polling design to minimize context switching, but consumes constant CPU cores even when idle.
- **Bdev (Block Device) Layer:** Manages disks using SPDK's bdev layer (supports aio and nvme drivers).
- **LVOL (Logical Volume) Store:** Stores data in SPDK logical volume stores, treating volumes as "blobs" for efficient snapshotting and thin provisioning.

Front-end Protocols

- **NVMe-oF (TCP):** Default frontend, exposing volumes as NVMe-over-Fabrics devices.
- **UBLK:** Alternative frontend using the Linux ublk framework for superior performance in high-spec environments.



V2 Performance Updates



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Environment

- OCI, Oracle Cloud Infrastructure
- VM.DenseIO.E5.Flex
- SAMSUNG MZWLR7T6HBLA-00AU3

FIO Parameters

- IOPS: bs=4K, iodepth=128, numjobs=8, norandommap=1
- Bandwidth: bs=128K, iodepth=16, numjobs=4
- Latency: bs=4k, iodepth=1

Baseline

- The baseline performance is measured with local-path-provisioner.

	IOPS (k)	Throughput (MiB/s)	Latency (us)
Read	1468	7115	79
Write	686	3895	24

V2 Performance Updates - Cont.



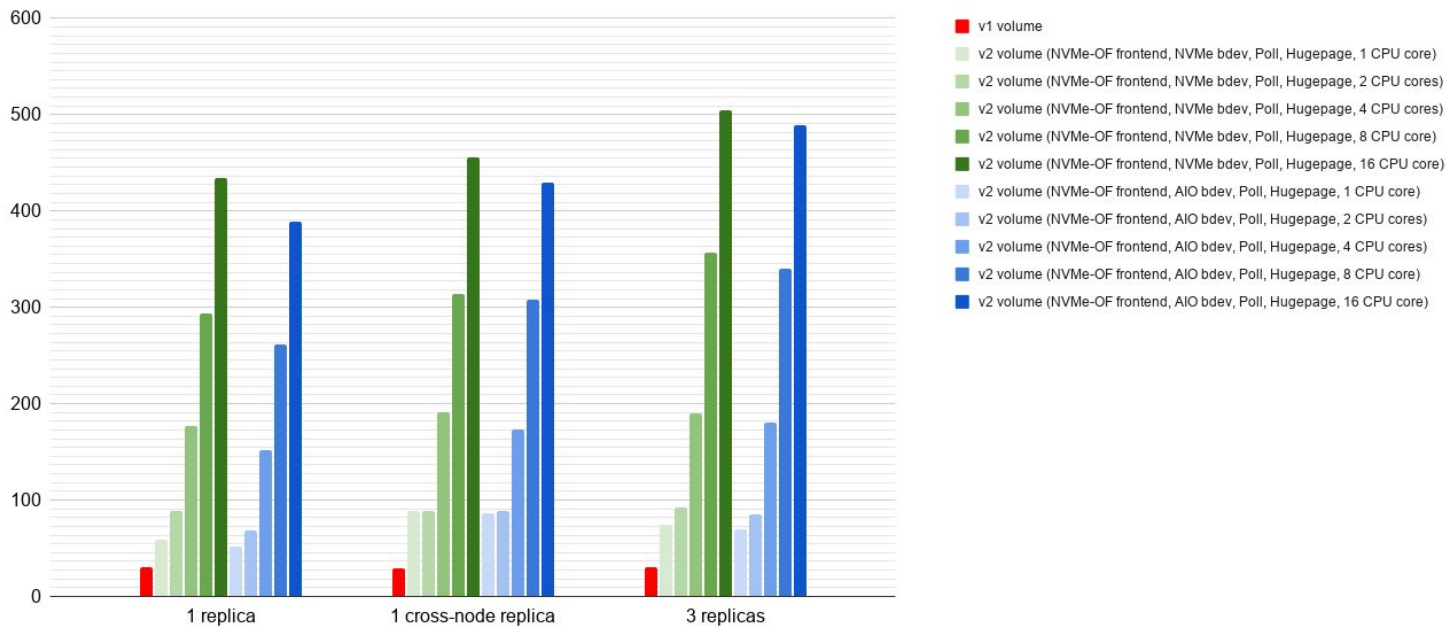
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Read IOPS (k)



V2 Performance Updates - Cont.



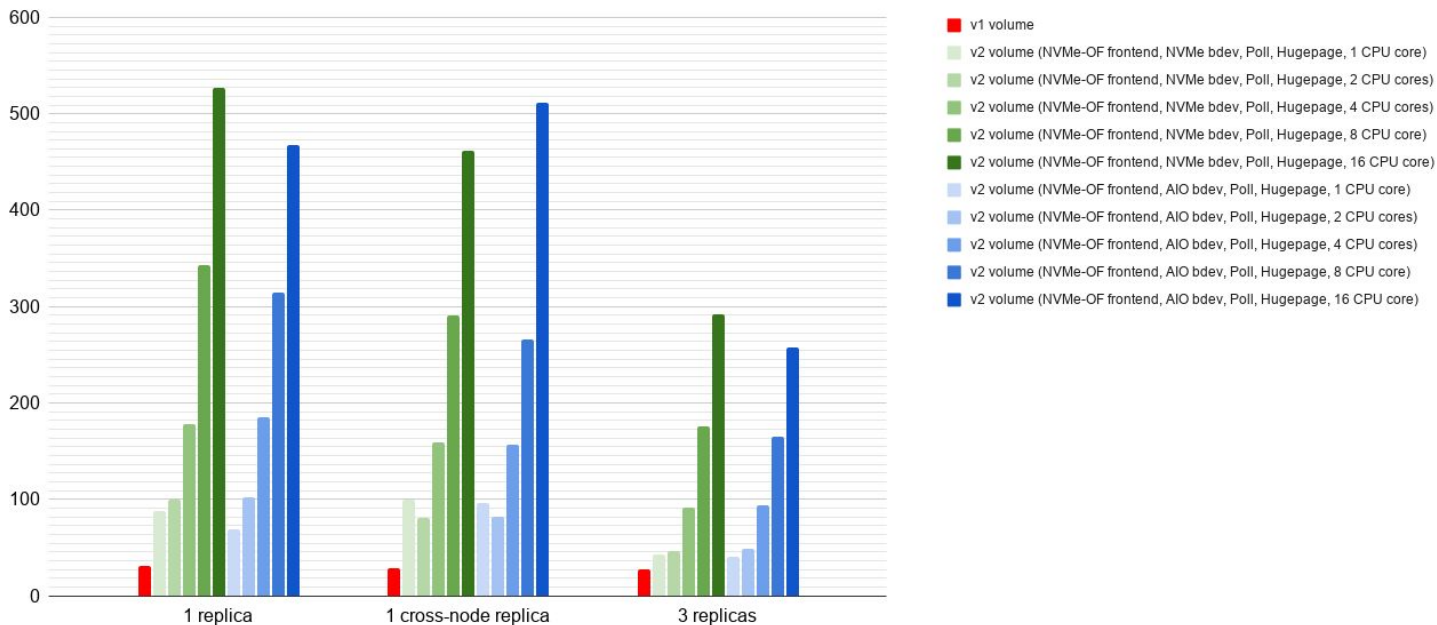
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Write IOPS (k)



V2 Performance Updates - Cont.



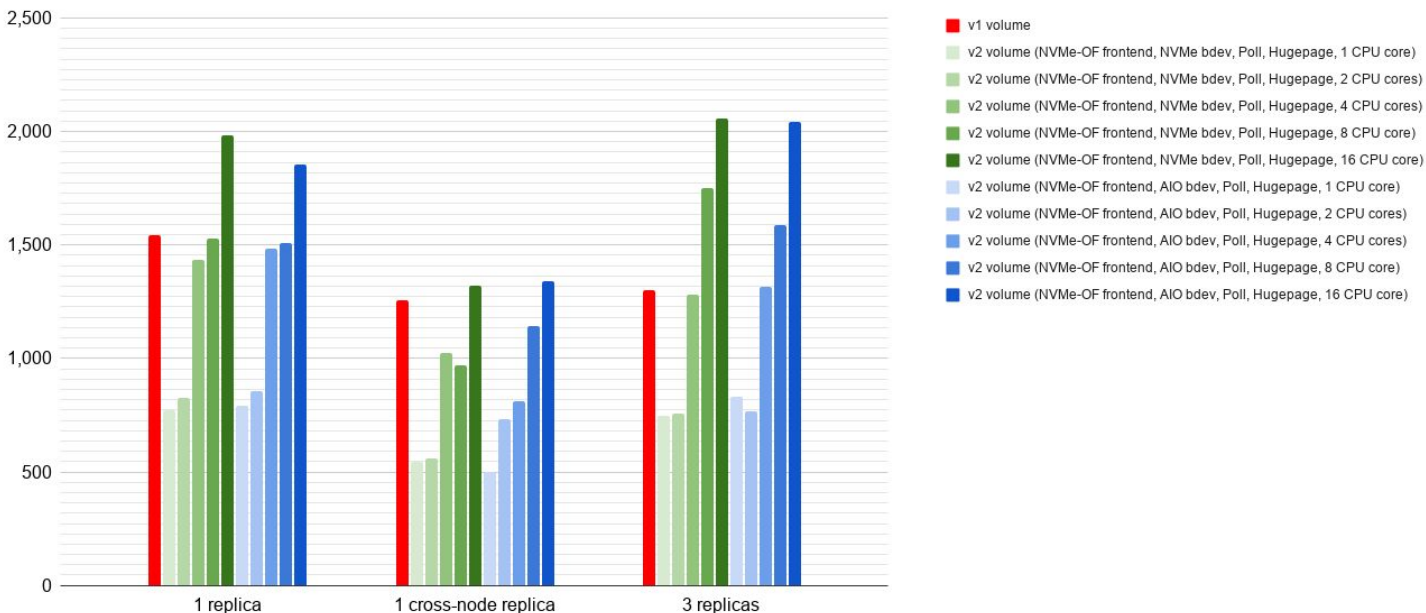
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Read Throughput (MiB/s)



V2 Performance Updates - Cont.



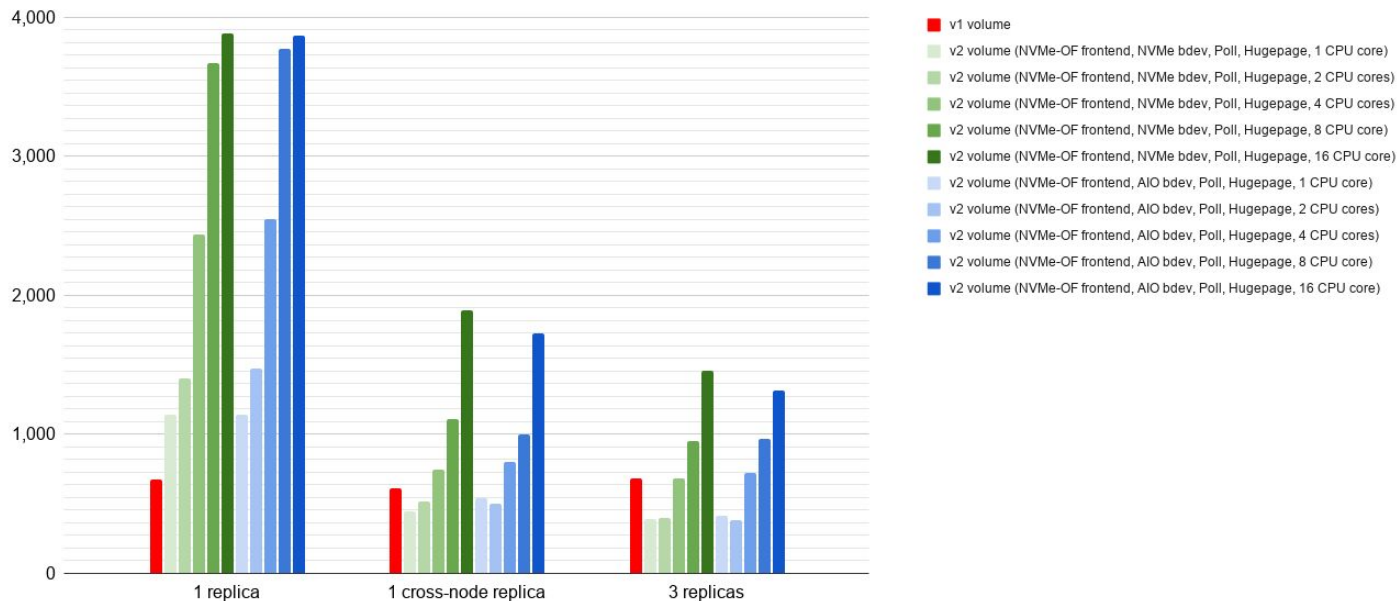
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Write Throughput (MiB/s)



V2 Performance Updates - Cont.



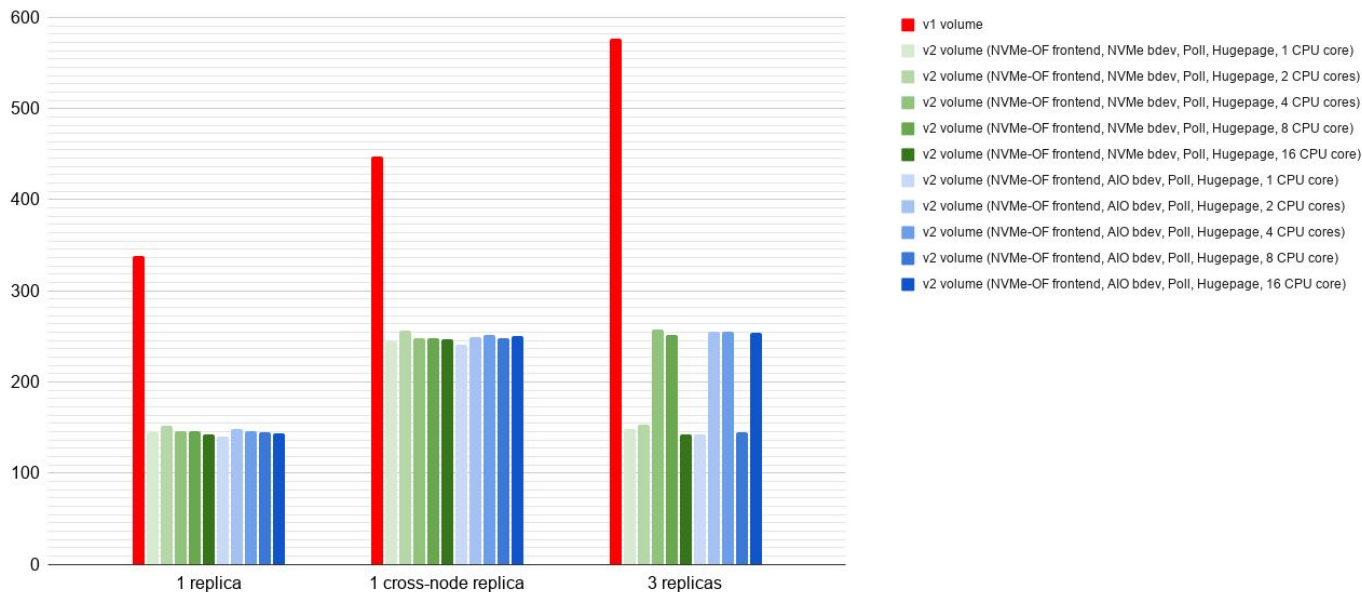
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Read Latency (microseconds)



V2 Performance Updates - Cont.



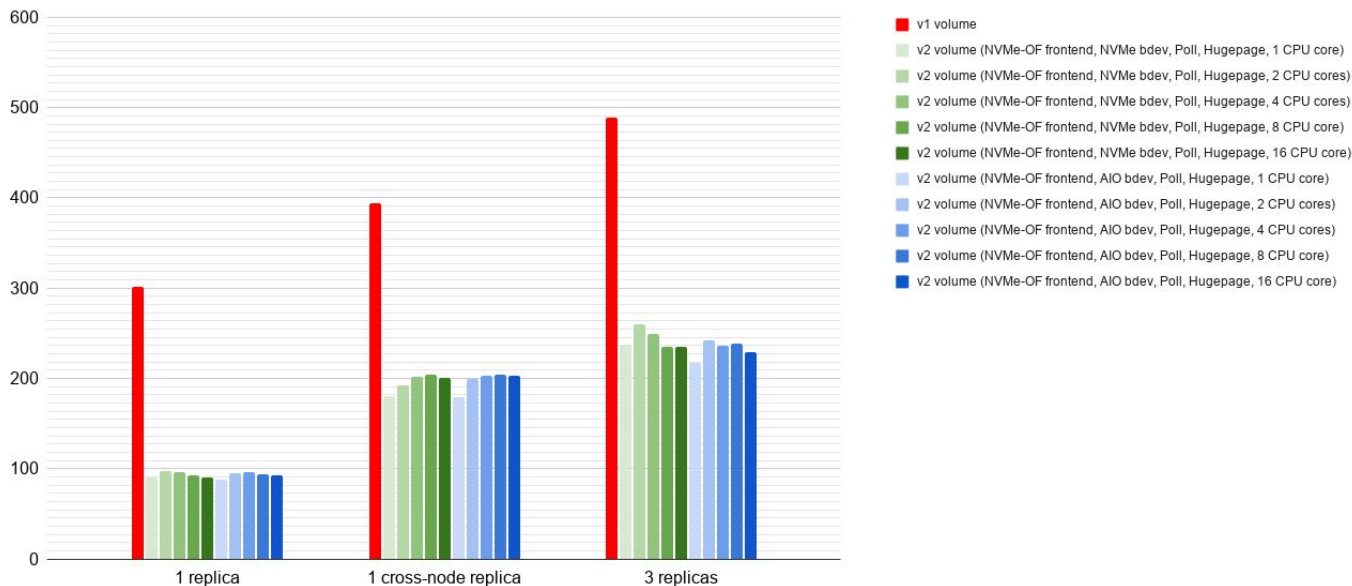
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Write Latency (microseconds)



1.11 Key Features

Outstanding Highlights - V2 maturity journey



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V2 Data Engine: Technical Preview

- Catching up with V1 features
- Increasing test coverage as V1

Features for V1 and V2					
Category	Feature	v1	v2	Doc	Note
Data Protection	Snapshot	✓	✓	link	
	Backup & Restore	✓	✓	link	
	DR Volume	✓	✓	link	
	System Backup & Restore	✓	✓	link	
	Snapshot Data Integrity Check	✓		link	Target version: TBU
RWX Volume	Creation/Deletion	✓	✓	link	
	Encryption	✓	✓		
	Migratable RWX Volume	✓	✓		
Volume Operations	Volume Expansion	✓	✓	link, v2 ticket	
	Volume Cloning	✓	✓	link, v2 ticket	
	Fast Volume Cloning	Not planned	✓	v2 ticket	
	Volume Encryption	✓	✓	link	
	Filesystem Trim	✓	✓	link	
Replica Scheduling	Replica Scheduling	✓	✓	link	

<https://github.com/longhorn/longhorn/wiki/V1-and-V2-Feature-Parities>

V2 Maturity - Test Coverage



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- In the automatic tests, v2 supports ~**62%** of v1's test cases (233/378), with 145 tests not yet enabled for v2.
- In the e2e suite (313), v2 has a dedicated test_v2.robot with 10 unique test cases, while 10 existing tests explicitly skip on v2 due to unsupported features (mainly engine image upgrades and some backing image/replica operations). ~**100%** coverage as V1.

● Here's a breakdown comparing v2 vs v1 test coverage across both test suites:

Manager Integration Tests (manager/integration/tests/)

Category	Count
Total test cases	378
Marked @pytest.mark.v2_volume_test (runs on v2)	233
v1-only (no v2 mark)	145

The v2_volume_test mark indicates a test has been validated to run against the v2 data engine. 145 tests (38%) are still v1-only.

e2e Robot Framework Tests (e2e/tests/)

Category	Count
Dedicated v2 file (test_v2.robot)	10
v2-only in other files (skip if DATA_ENGINE == v1)	1 (in test_basic.robot)
v1-only in other files (skip if DATA_ENGINE == v2)	10
Shared (run on both)	majority

The v1-only skips are spread across:

- test_engine_image.robot: 3 (engine image upgrade not supported on v2)
- test_replica.robot: 2
- test_backing_image.robot: 2
- test_backup.robot: 1
- test_settings.robot: 1
- test_basic.robot: 1

● **313 total test cases** across 63 robot files, broken down by directory:

Directory	Files	Test Cases
e2e/tests/negative/	29	~185
e2e/tests/regression/	30	~124
e2e/tests/ (root)	4	4
Total	63	313

Notable files by count:

- negative/node_reboot.robot: 28
- negative/ha_volume_migration.robot: 23
- negative/kubelet_restart.robot: 13
- regression/test_basic.robot: 12
- regression/test_backing_image.robot: 12
- regression/test_scheduling.robot: 12
- regression/test_v2.robot: 10 (v2-dedicated)

1.11 Key Features - Cont.

Outstanding Highlights - replica building and scheduling efficiency



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Support for ublk Frontend with Performance Tuning Parameters

- ublk (Userspace Block Device), introduced in 1.9 for Kernel v6.0+ environments as an alternative to NVMe-oF for V2 Data Engine volumes, now includes performance tuning parameters in 1.11 to adjust queue size and number of queues, improving IO performance.

Faster Replica Rebuilding from Multiple Sources

- The V1 Data Engine now supports parallel rebuilding, streaming data from multiple healthy replicas simultaneously instead of a single source. This significantly reduces the time to restore redundancy for volumes with scattered data chunks.

1.11 Key Features - Cont.

Outstanding Highlights - replica building and scheduling efficiency



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Balance-Aware Algorithm Disk Selection For Replica Scheduling

- Longhorn improves the disk selection for the replica scheduling by introducing an intelligent balance-aware scheduling algorithm, reducing uneven storage usage across nodes and disks.

StorageClass allowedTopologies Support

- Administrators can now use the allowedTopologies field in Longhorn StorageClasses to restrict volume provisioning to specific zones, regions, or nodes within the cluster.

1.11 Key Features - Cont.

Outstanding Highlights - monitoring, storage network segregation, RWOP



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Node Disk Health Monitoring

- Longhorn now actively monitors the physical health of the underlying disks used for storage by using S.M.A.R.T. data. This allows administrators to identify issues and raise alerts when abnormal SMART metrics are detected, helping prevent failed volumes.

Share Manager Networking

- Users can now configure an extra network interface for the Share Manager to support complex network segmentation requirements.

ReadWriteOncePod (RWOP) Support

- Access mode that restricts volume access to exactly one pod in the entire cluster.

1.12 Roadmap

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V2 Data Engine New Features

- General Availability
- *Volume Live Upgrade (1.12 to 1.13)
- IPv6 Support
- Engine Migration: Cross-Node NVMe-oF Initiator and Target
- Orphan Resource Cleanup
- Fast Clone

New Features

- In-Place Pod Resource Resize for Instance Manager
- Offline Replica Rebuilding by default
- Replica Rebuilding Resume at runtime

Experimental

- *Sharding Mode Volume (POC) for V2



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Community Highlights



Agenda

- Current status of the Longhorn community
 - The numbers
 - What's new
- Future roadmap

The numbers

- Slack
 - CNCF Slack: 1950
 - Rancher Users Slack: 2339
- YouTube: 430

Contributor engagement (v1.11.0)



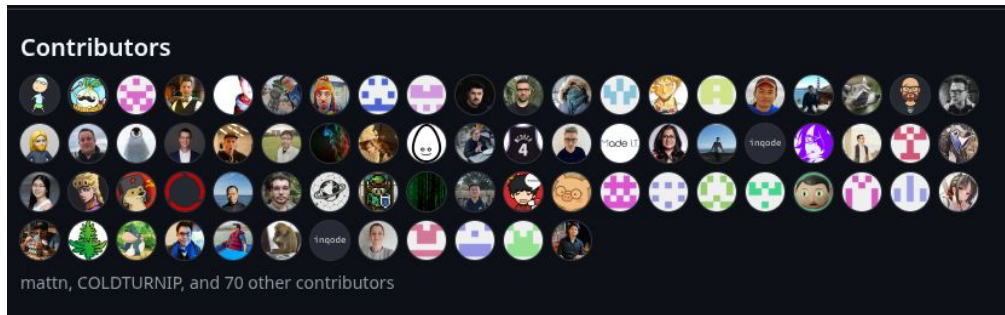
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- **72 global** contributors
 - ~**50%** were non-SUSE contributors
 - Most popular ways of contributing to the release:
 - Bug reports
 - Feature requests
 - Code contributions



Bonus!



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February 4, 2 PM UTC

Global Online Meetup: Longhorn

Hosted by Divya Mohan ft special guest,
David Ko



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Monthly community meetings

- Alternate between APAC and AMER/EU-friendly
- Recordings on official YouTube, whenever meetings are in quorum
- APAC-friendly: **6 AM UTC** (previous 8 AM UTC)
- AMER-friendly: **4 PM UTC**
- [Agenda doc](#)
 - For edit access, please join the mailing lists in the agenda doc

- New ways to stay updated with the Longhorn project
 - [LinkedIn](#)
 - [Twitter](#)
 - [Bluesky](#)

Share your stories with us!



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hookak and derekbit docs(adopters): add NAVER as an end-user adopter

d8e7254 · yesterday History

Preview Code Blame 17 lines (14 loc) · 1.13 KB ·

Raw

Note

To encourage adoption of Longhorn and share your experience with the community, please add your information below.

Adopter Types

- End-User: An organization that runs Longhorn in a production environment.
- Integration: An organization offering a product that integrates with Longhorn by providing additional services or functionality.
- Vendor: An organization that includes Longhorn as part of its packaged product.
- Hosted-Service: An organization that operates Longhorn within its hosted service, offered as a product to customers.

Longhorn Adopters

Type	Name	Website	Use-Case
End-User	Nemlig	https://www.nemlig.com/	A comprehensive, persistent storage solution on Kubernetes. Part of "Kubernemlig," Nemlig's internal, developer- and operations-friendly Platform-as-a-Service (PaaS).
End-User	Child Rescue Coalition	https://childrescuecoalition.org/	Kubernetes persistent replicated storage with automated backups.
End-User	NAVER	https://www.navercorp.com/	Production Kubernetes persistent storage supporting NAVER's internal platform and cloud-native services at scale.

Future roadmap

- Theme for this year: Improving project health
 - Building our governance structure
 - Identifying & dividing project into areas & roles for contribution
 - Devising a community ladder for transparency
 - Broadening our contributor pool - steady & drive-by
 - Growing existing contributors into maintainers

If any of this interests **you**, let's talk!

Stretch goals

- Participation in ecosystem-wide programs (LFX Mentorship etc.)
- Cross-ecosystem collaboration



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Q & A

Thank you!

